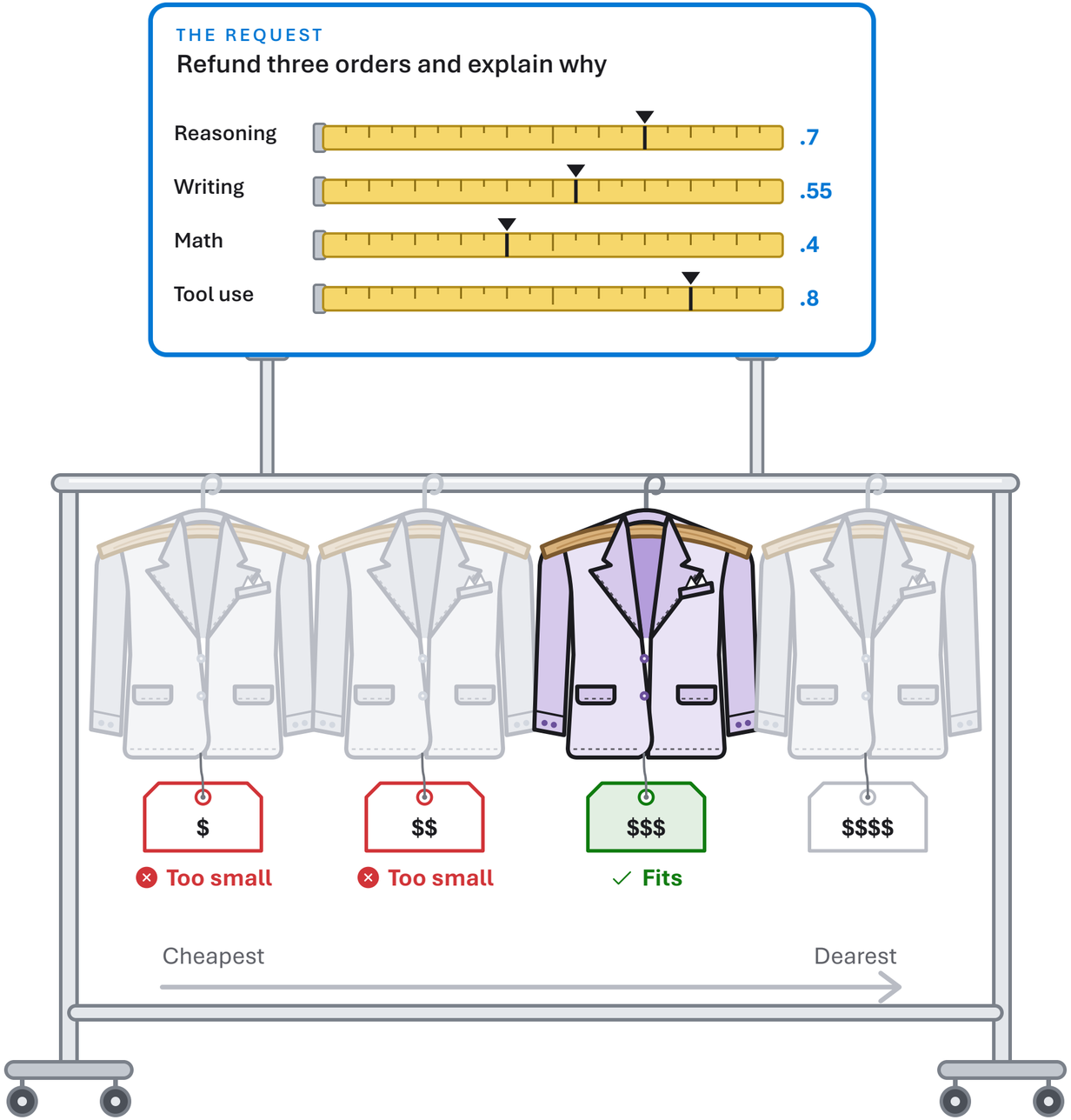
Microsoft

CONTOSO SUPPORT · STUDY

The cheapest model that fits each request

How the assistant picks a model, and what it saved.

Sample data



Sources: Sample data, made up for this example

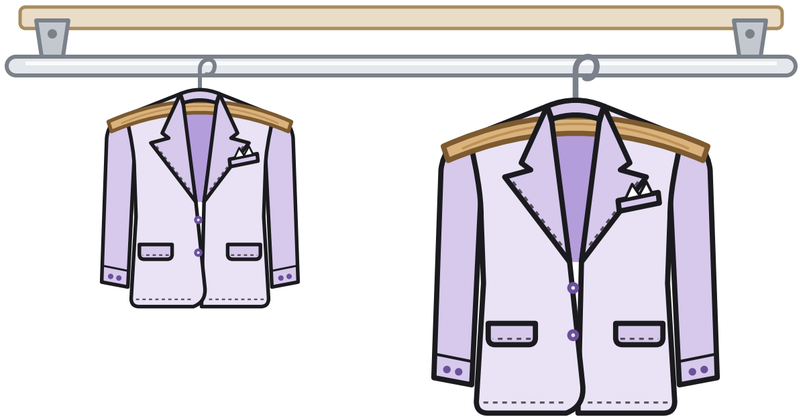
WHERE WE STARTED

The old router knew only big or small

OLD ROUTER · 2025



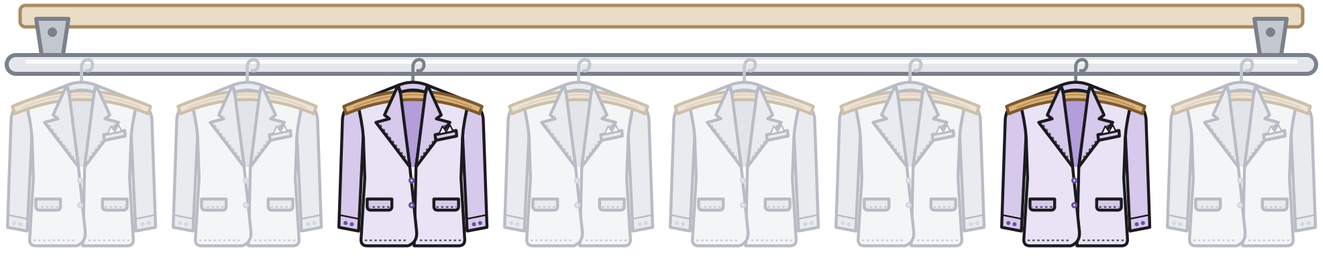
One score:
big or small



Kite S
SMALL

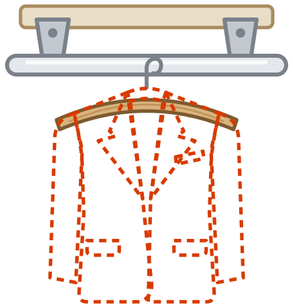
Heron L
BIG

MODEL POOL · 8 MODELS



Wren S · Wren M · **Kite S** · Kite M · Kite L · Heron M · **Heron L** · Heron XL

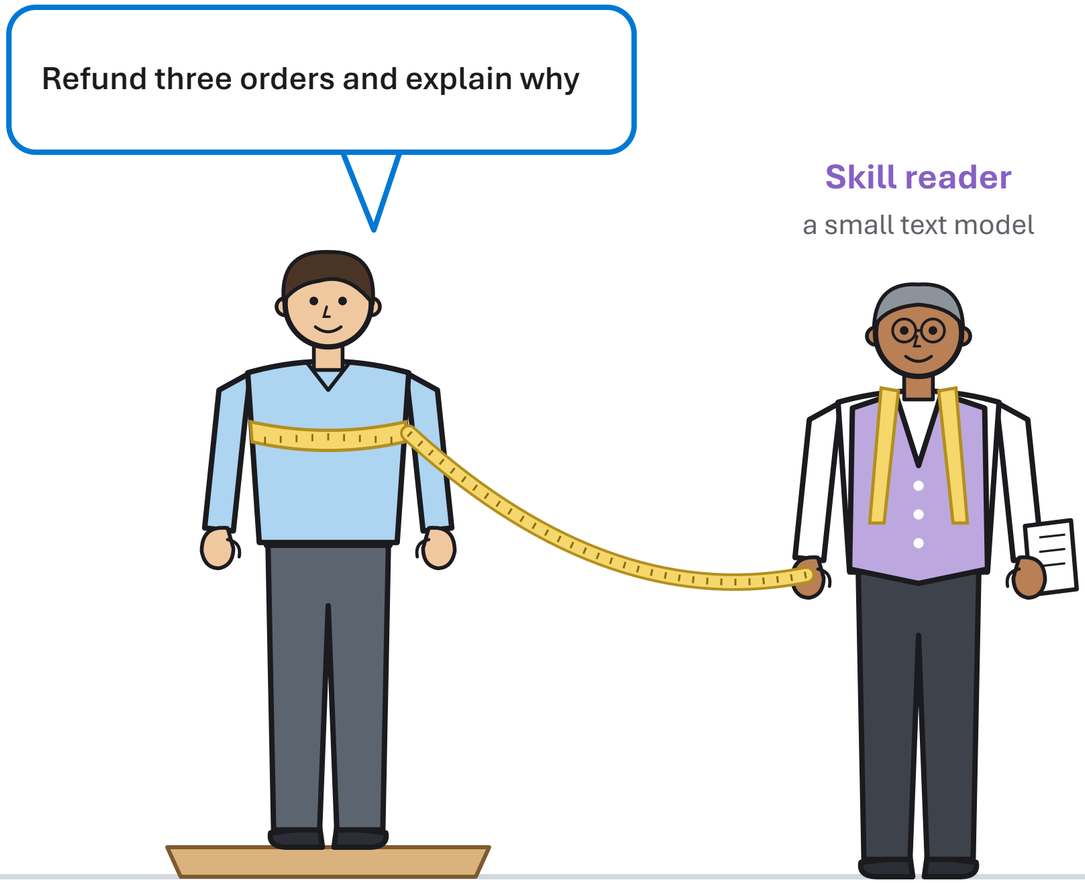
It uses two of the eight



New model? Retrain.

HOW IT WORKS · 1

First, measure what the request needs

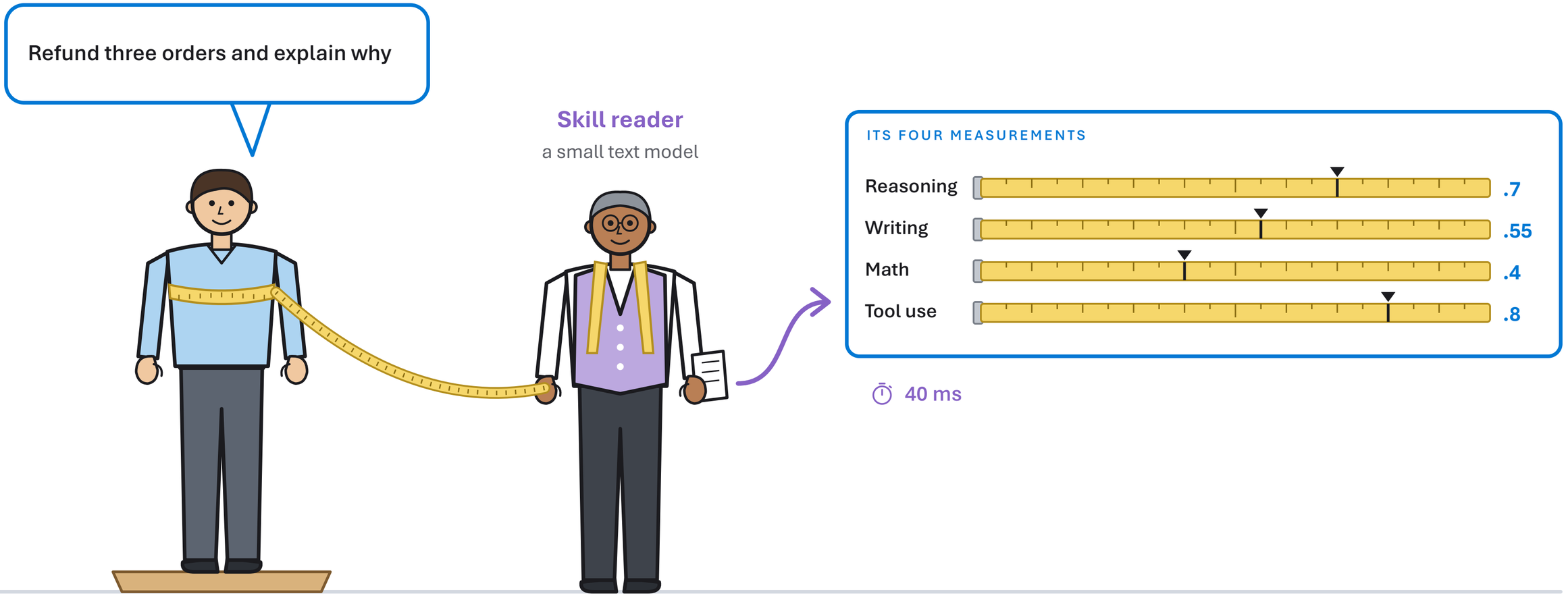


1 2 Measured by a small model

Sources: Sample data, made up for this example

HOW IT WORKS · 1

First, measure what the request needs

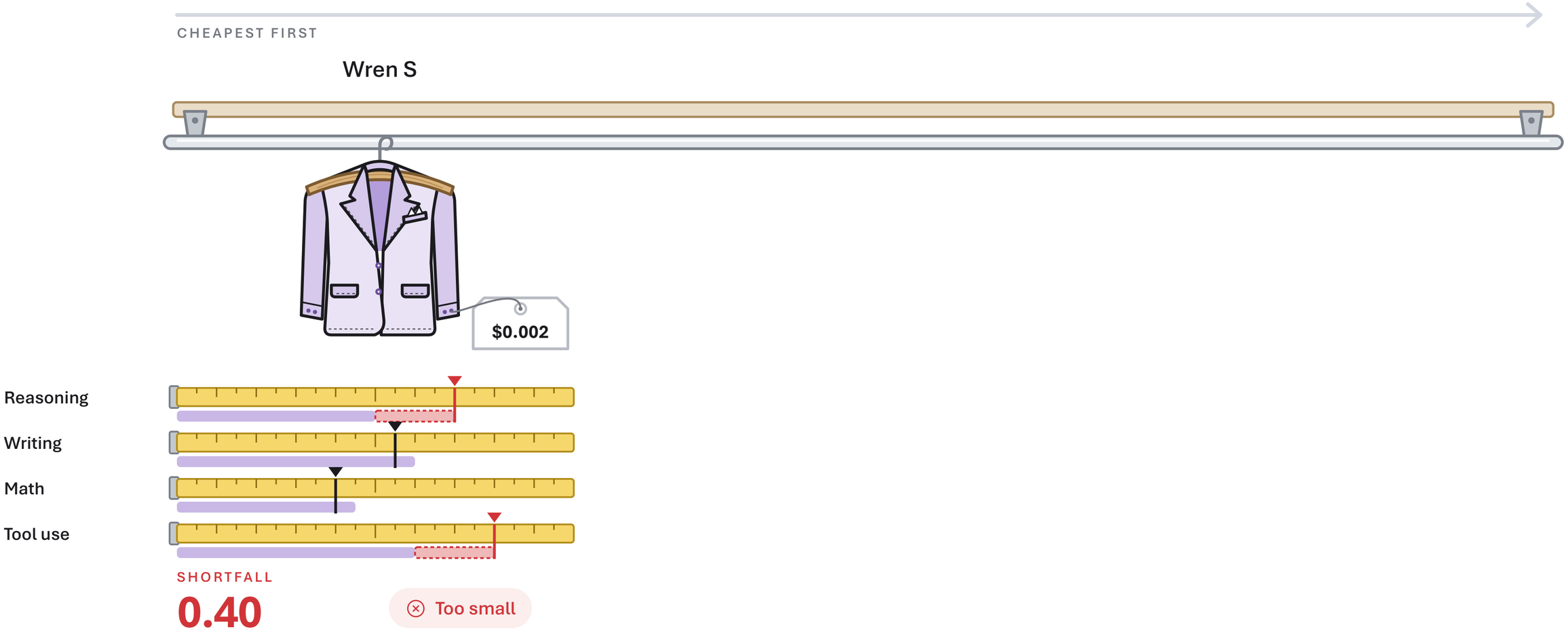


1 2 Out come four needs

Sources: Sample data, made up for this example

HOW IT WORKS · 2

The cheapest model that fits every need wins

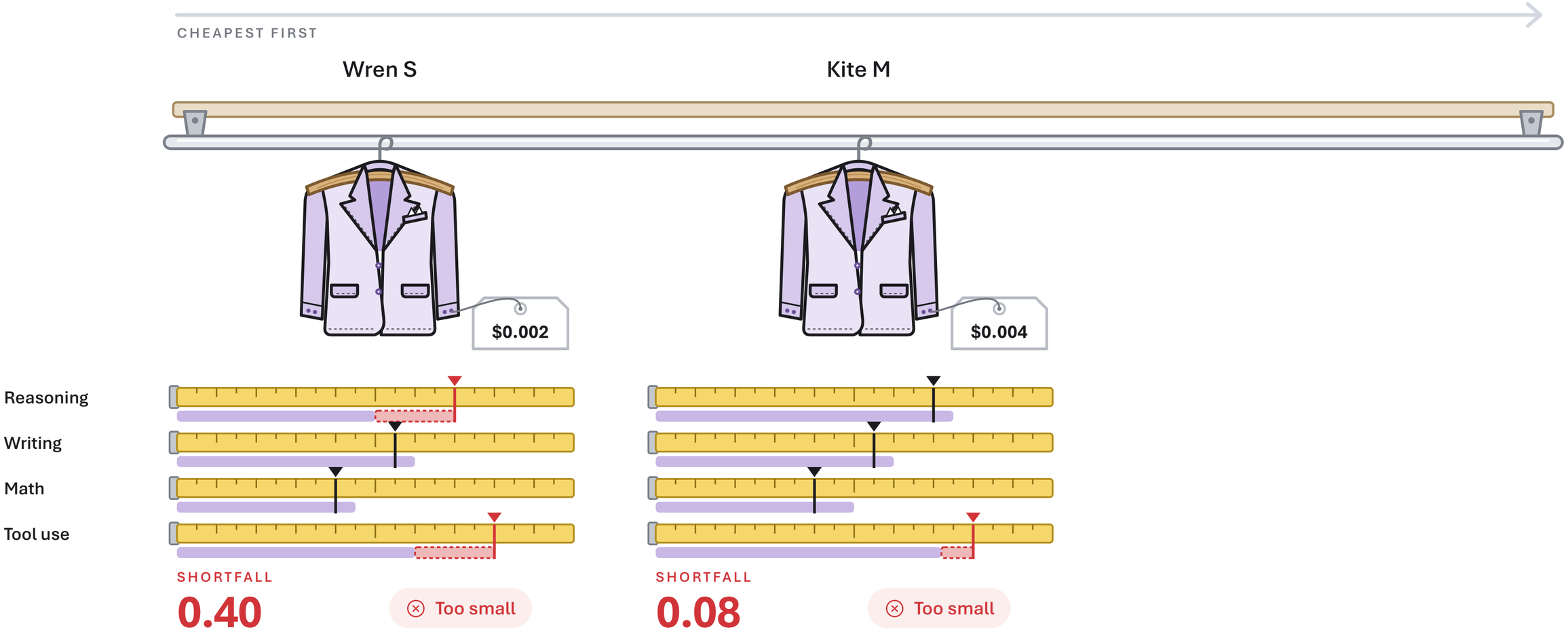


1 2 3 **Wren S: short by 0.40**

Sources: Sample data, made up for this example

HOW IT WORKS · 2

The cheapest model that fits every need wins

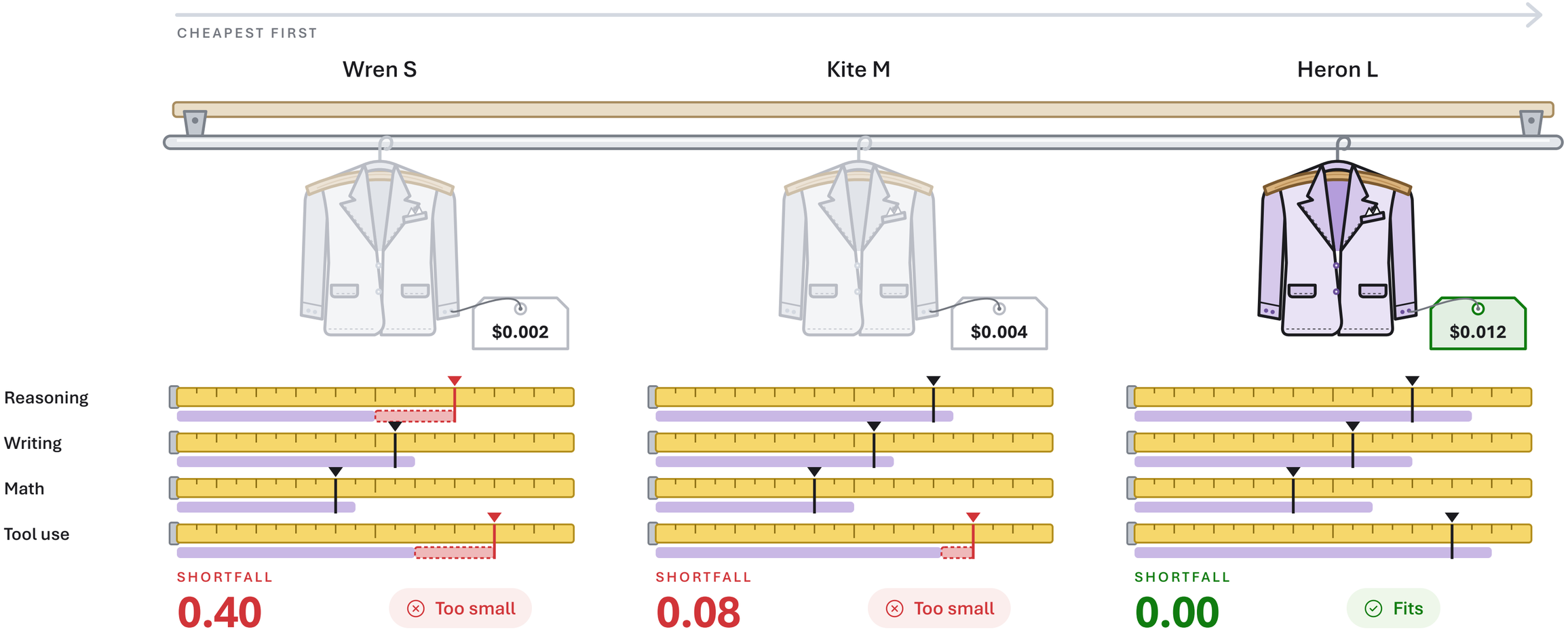


1 2 3 Kite M: short by 0.08

Sources: Sample data, made up for this example

HOW IT WORKS · 2

The cheapest model that fits every need wins



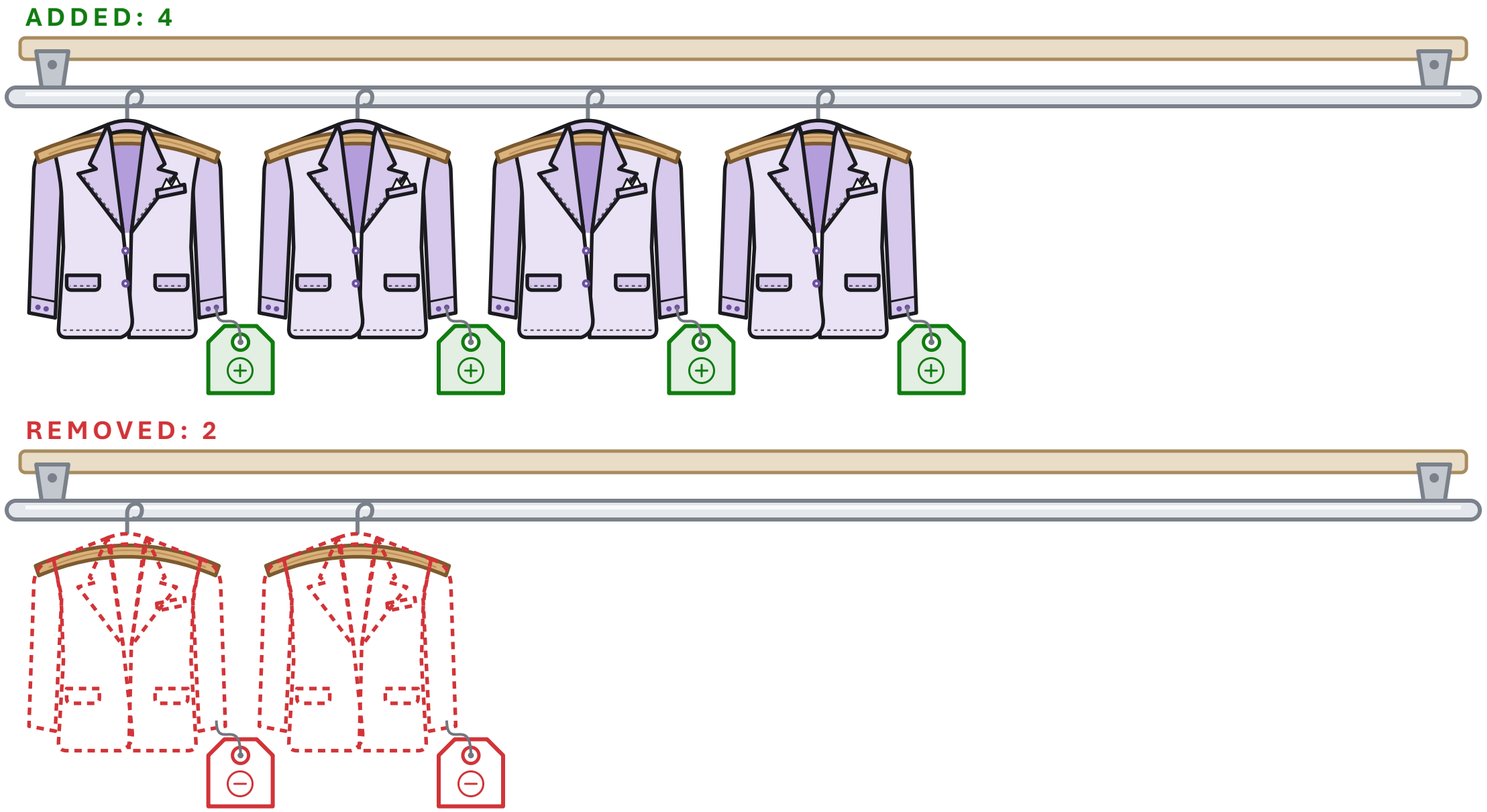
1 2 3 **Heron L fits every need, so it wins**

Sources: Sample data, made up for this example

THE MODEL LIST

New models join with a config edit

`models.yaml`



0

retraining runs

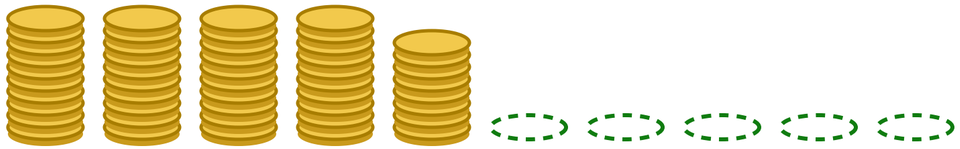
over three months

A new model gets traffic on day one

Sources: Sample data, made up for this example

RESULTS · 1,000 TICKETS

Same quality for under half the cost

	COST · EACH STACK IS 10% OF ALWAYS-HERON XL	SAVED	QUALITY
Always Heron XL the biggest model		0%	90.2
Old router big or small		12%	89.5
Fitting **the cheapest that fits**		**52%**	**89.9**

Sources: Sample data, made up for this example

ACROSS LANGUAGES

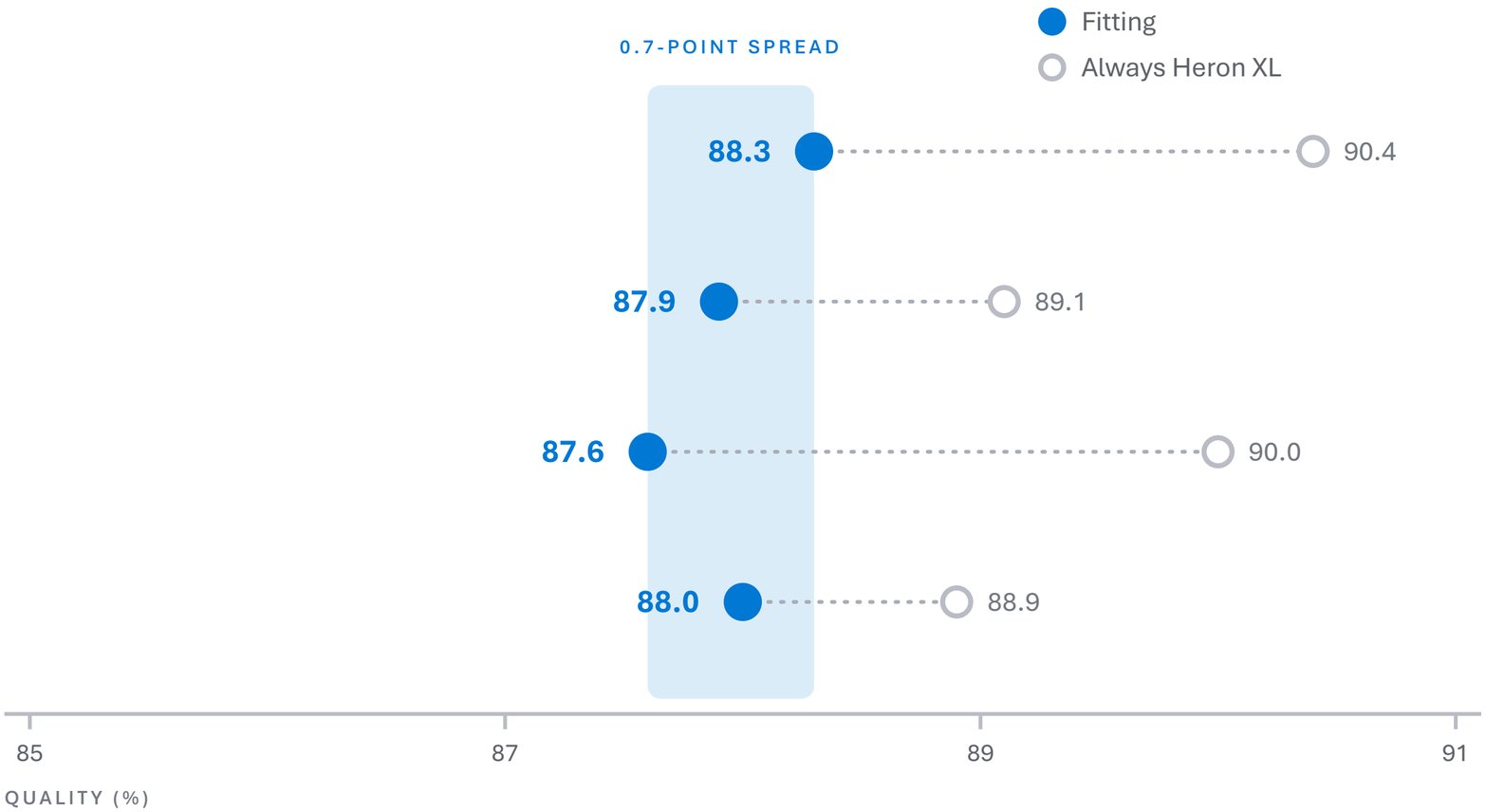
Steady across twelve languages

English
600 tickets

European
fr de es it · 240

East Asian
zh ja ko · 90

Other
ru ar tr vi · 70

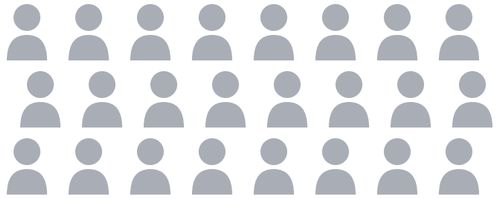


Sources: Sample data, made up for this example

A/B TEST

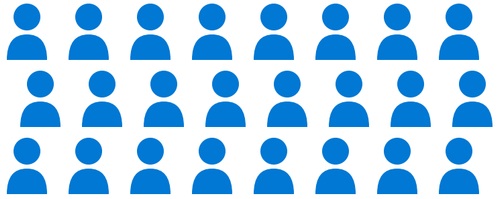
A two-week test: faster, fewer errors

CONTROL · OLD ROUTER



12K users

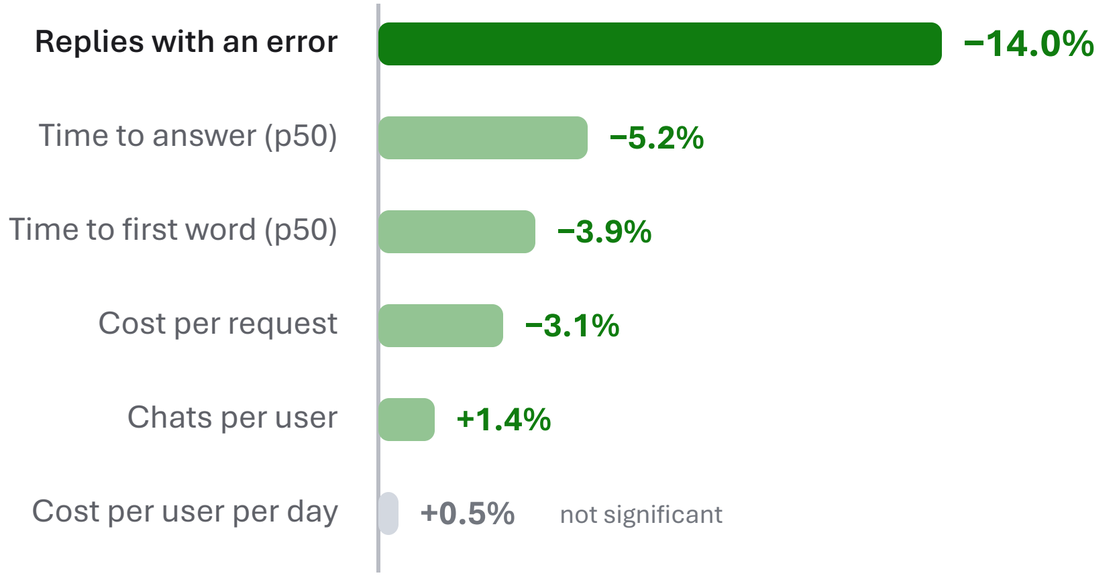
FITTING



12K users

14 DAYS · 2–15 SEP 2026 · SUPPORT CHAT

CHANGE WITH FITTING · GREEN IS BETTER



ⓘ Thumbs-down +3%: mostly the assistant’s edits, not the pick

Sources: Sample data, made up for this example

IN PRODUCTION

Live for every Contoso support chat

Support chat

Contoso

410K

requests a week



Fitting

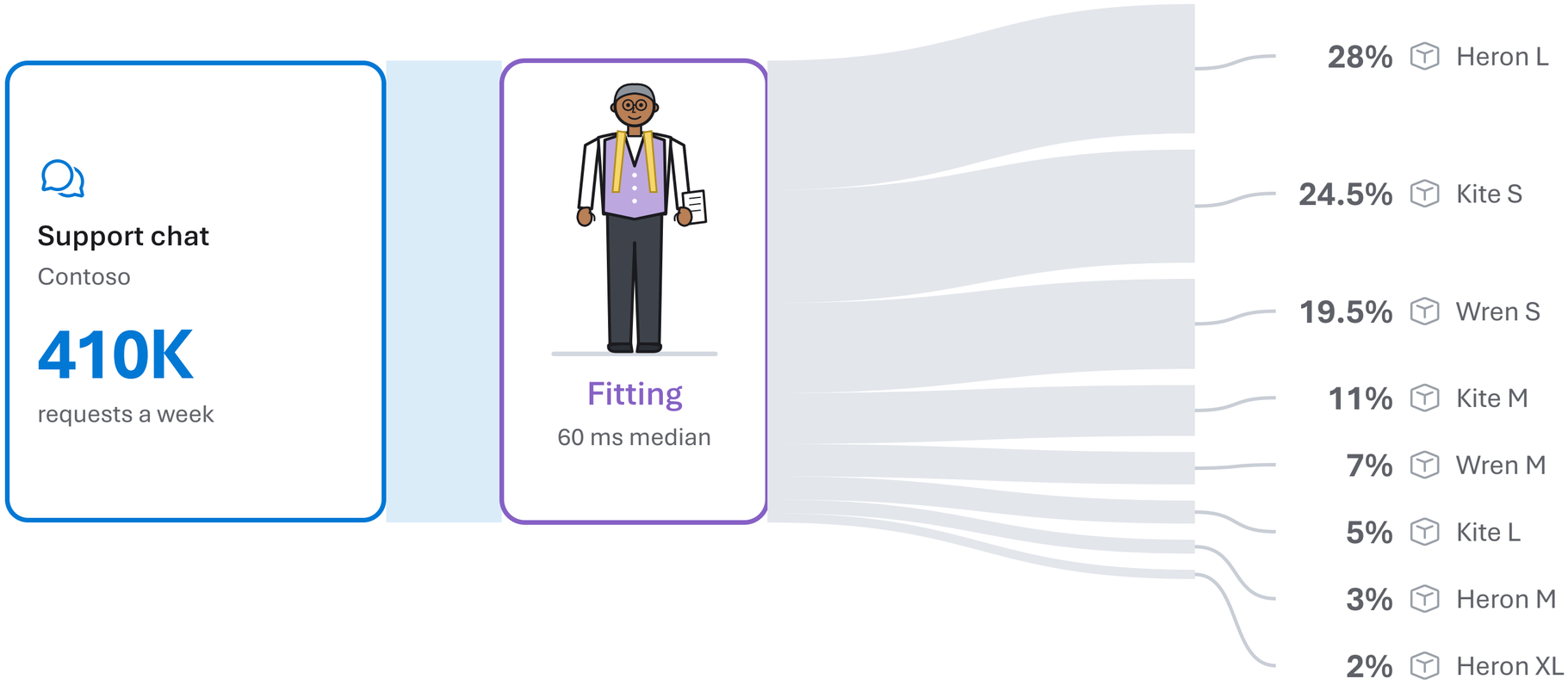
60 ms median

1 2 3 410K requests a week

Sources: Sample data, made up for this example

IN PRODUCTION

Live for every Contoso support chat

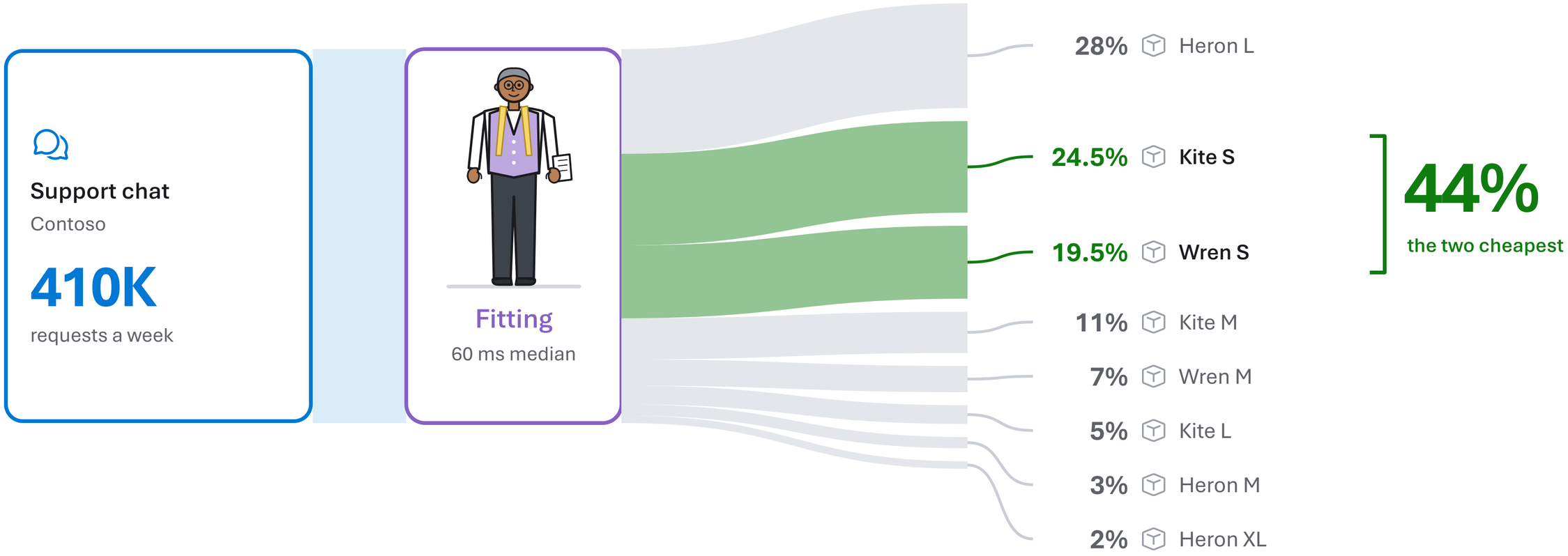


1 2 3 Each model gets a share

Sources: Sample data, made up for this example

IN PRODUCTION

Live for every Contoso support chat



1 2 3 The two cheapest: 44%

Sources: Sample data, made up for this example

WHAT COMES NEXT

Next: images, agents and long chats

